

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

School of Computing — Department of Computer Science & Engineering

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Course: 21CSC203P — Advanced Programming Practice	Department: Computer Science and Engineering

Unit III | Session 1 | SLO 1: Introduction to Concurrent Programming (Code: 3011)

Q: 1. Fill in the blank:

In concurrent programming, multiple _____ can be executed in overlapping time periods.

Ans: Answer: threads (or processes / tasks)

Q: 2. True or False:

A thread is a lightweight process used in concurrent programming.

Ans: Answer: True

Q: 3. Multiple Choice:

Which of the following is commonly used for managing concurrency?

A) Semaphore B) Compiler C) Keyboard D) Monitor

Ans: Answer: A) Semaphore (and D is also a concurrency construct; Semaphore is the standard synchronization primitive).

Q: 4. Fill in the blank:

A _____ condition occurs when two or more threads wait for each other indefinitely.

Ans: Answer: deadlock

Q: 5. True or False:

Concurrency and parallelism refer to the exact same concept.

Ans: Answer: False (Concurrency is dealing with lots of things at once; parallelism is doing lots of things at once simultaneously).

Q: 6. Fill in the blank:

A _____ occurs when two threads simultaneously modify shared data leading to inconsistent states.

Ans: Answer: race condition

Q: 7. True or False:

Threads belonging to the same process share the same memory space.

Ans: Answer: True

Q: 8. Fill in the blank:

Ensuring that multiple threads execute coordinated access to shared variables is called thread _____.

Ans: Answer: synchronization

Q: 9. Multiple Choice:

A mutual exclusion mechanism that allows only one thread into a critical section is called:

A) Mutex B) JVM C) Garbage Collector D) Compiler

Ans: Answer: A) Mutex

Q: 10. Fill in the blank:

A _____ ensures that only one thread accesses a critical section at a time.

Ans: Answer: mutex (or lock / monitor)